Computer Networks

Homework 1 (CHAP. 1)

1. Suppose users share a 1 Mbps link. Also suppose each user requires 100 kbps when transmitting, but each user transmits only 10 percent of the time. (See the discussion of statistical multiplexing in Section 1.3.)
   a. When circuit switching is used, how many users can be supported?
   b. For the remainder of this problem, suppose packet switching is used. Find the probability that a given user is transmitting.
   c. Suppose there are 40 users. Find the probability that at any given time, exactly n users are transmitting simultaneously. (Hint: Use the binomial distribution.)
   d. Find the probability that there are 11 or more users transmitting simultaneously.

2. What are the differences between circuit switching and packet switching? What are the advantages and disadvantages for packet switching, compared with circuit switching?

3. Observe what is a possible packet routing path from a country outside Asia (e.g. US or any EU country) to our school. You can follow following steps to do the observation.
   Step 1. Find a free visual traceroute web site from internet.
   Step 2. Usually internet free visual traceroute websites would provide an option to let you enter the destination (IP or URL) you would like to view, in which put our school website (www.ntou.edu.tw) to view the traceroute result.

   a. what is the visual website you used?
   b. what is the path you got?
   c. check the global submarine cable map from internet (e.g. https://www.submarinecablemap.com/). Referring to the submarine cable map for those cable connecting to Taiwan and the path you got above, Guess which submarine cables are utilized for the traceroute results according to the path you observed, Why?

4. What is the internet access technology you use at home (e.g. ADSL, cable modem, …)? What is the maximal bandwidth your internet access ISP announced? Do a bandwidth testing (you can find many tools/services for internet speed test), and check do your internet connect can reach the maximal bandwidth as your ISP claimed? If not, what is the possible reasons?

5. Please check with some job hunting/matching websites (e.g. 104, or 1111) and find three companies related in network/telecommunication technique. What kinds of network related knowledge are preferred in these companies?